

Long-Context

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Long-Context Modelling

What:

Process, retain, and reason over **very long input sequences**, ranging from **tens of thousands to millions of tokens**

Why:

Many real-world tasks exceed the context length of early transformers (2k–4k tokens):

- Document-level and multi-document QA
- Code understanding across large repositories
- Legal, financial, and scientific analysis
- Tool-using agents with long interaction histories

Without long-context modeling, systems rely on **retrieval, truncation, or summarization**, which can lose critical information.

Core Challenges

Attention Degradation

- Even when long contexts are supported, models may:
 - Ignore early tokens (“recency bias”)
 - Fail to retrieve relevant information
 - Overweight irrelevant or recent content
- Long context $\neq$ effective long-range reasoning.

Training Data Mismatch

- Most pretraining data consists of **short documents**.
- Models extrapolating to long contexts face:
 - Distribution shift
 - Positional generalization failures

Computational Cost

- Self-attention scales quadratically with sequence length
- This makes naive scaling prohibitively expensive in memory and compute.

Outline

- Positional Encodings
- Efficient Attention
- Evaluation
- Analysis
- Summary

Positional Encodings

Positional Encodings: Recap

Attention Is All You Need/Transformer

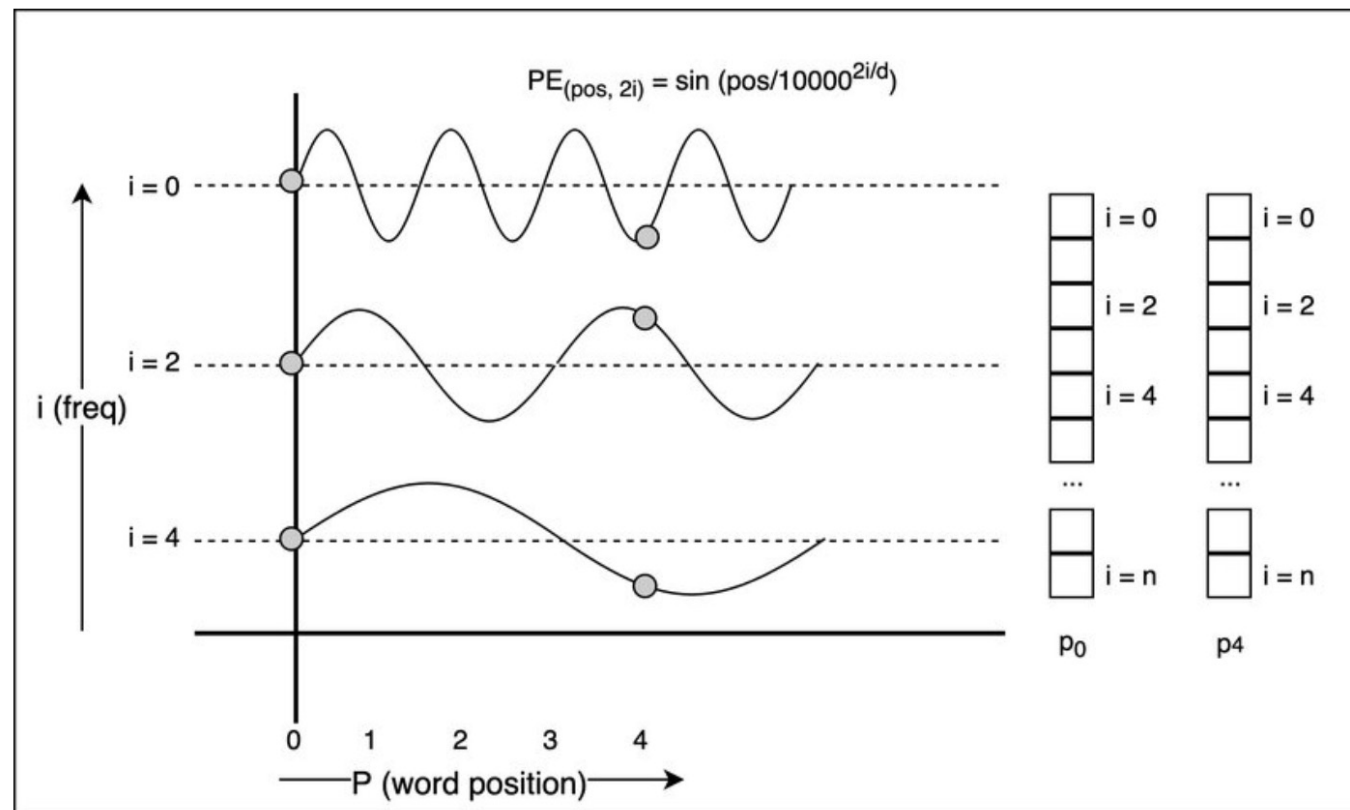
- Order of elements is crucial – word embeddings do not contain position
- Transformer processes all tokens in parallel
- Token embeddings x_{pos} + Positional encodings $PE_{pos} \rightarrow$ Transformer attention layers

$$z_{pos} = x_{pos} + PE_{pos}$$

Positional Encodings: Sinosoidal

Sinusoidal positional encoding are fixed

- Low frequency (high i) $\rightarrow$ slow oscillations (global position)
- High frequency (low i) $\rightarrow$ fast oscillations (local position)



[Researchgate.net](https://www.researchgate.net)

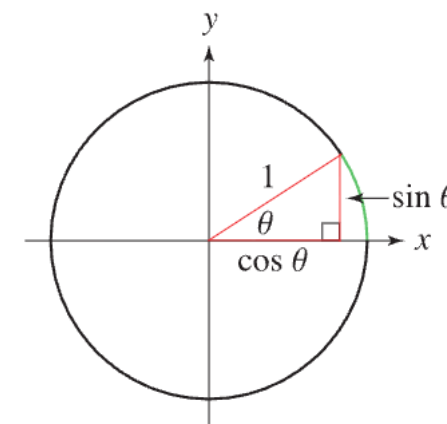
Positional Encodings: Sinosoidal

$$\text{PE}_{\text{pos}} = [\sin(\omega_0 \text{pos}), \cos(\omega_0 \text{pos}), \sin(\omega_1 \text{pos}), \cos(\omega_1 \text{pos}), \dots, \sin(\omega_{d/2-1} \text{pos}), \cos(\omega_{d/2-1} \text{pos})]$$

Where $\omega_i = \frac{1}{10000^{2i/d}}$, d = transformer embedding dim

- **Properties**

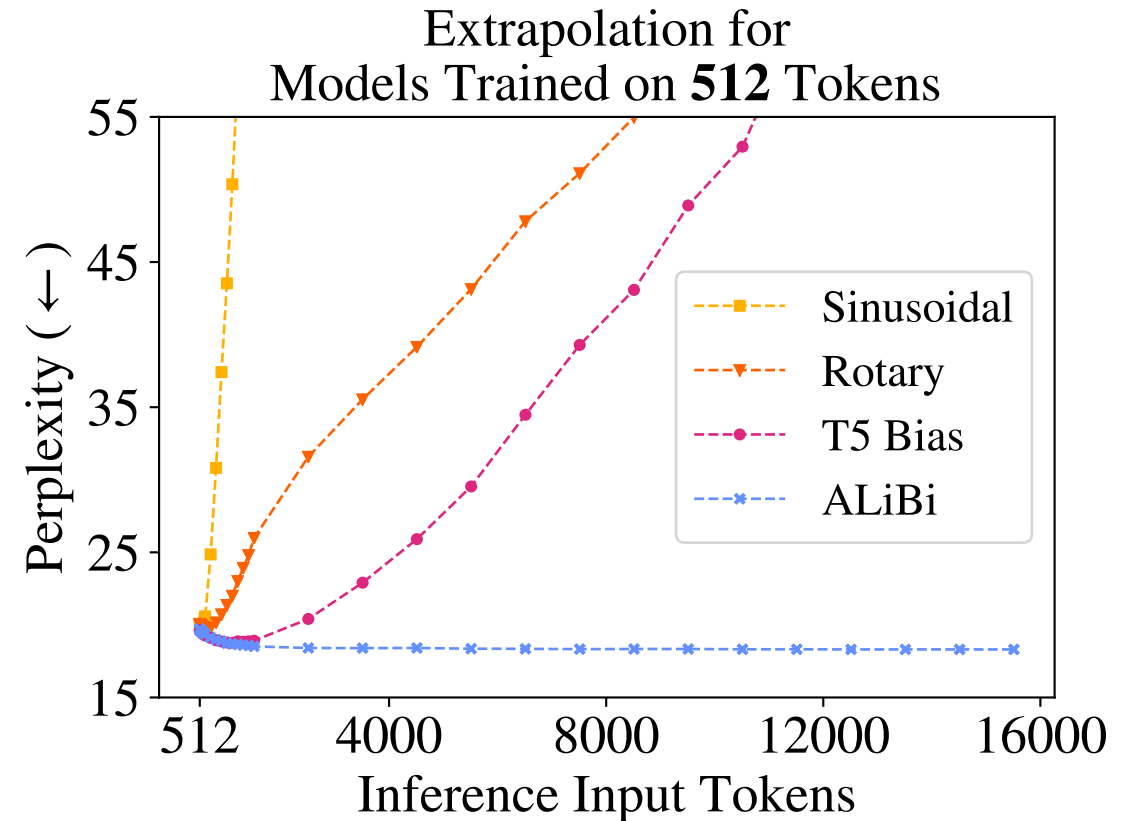
- sin cos pair defines a **single angle** on a unit circle
- PE vector forms a **unique fingerprint** for each position
- Can infer how far apart two tokens or pos are: dot product of two PE contain **relative distance**
- Deterministic (no learning)
- **Extrapolate** to longer sequences than seen in training



Positional Encoding Design

Sinusoidal PE does not extrapolate well beyond training length

Rotary methods are more promising



Press et al. (2021) [Train Short, Test Long: Attention with Linear Biases Enables Input Length Extrapolation](#)

RoPE

Addition (sinusoidal PE)

- Position is *attached* to the vector
- Relative distance must be *learned*

Rotation (RoPE)

- Position *transforms* the vector
- Turns absolute position → angle
- Turns relative position → angle difference so it is **built into the math** no learning required.

Su et al. (2021) [Roformer: Enhanced transformer with rotary position embedding](#)

RoPE

2D Example:

Query $q = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, **Key** $k = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Attention without position: $q \cdot k = 1$

Rotation matrix R for position pos : $\begin{bmatrix} \cos(pos) & -\sin(pos) \\ \sin(pos) & \cos(pos) \end{bmatrix}$

Query at position 1: $q_1 = R(1)q = \begin{bmatrix} \cos 1 \\ \sin 1 \end{bmatrix} \approx \begin{bmatrix} 0.54 \\ 0.84 \end{bmatrix}$ **Rotary position embedding**

Key at position 3: $k_3 = R(3)k = \begin{bmatrix} \cos 3 \\ \sin 3 \end{bmatrix} \approx \begin{bmatrix} -0.99 \\ 0.14 \end{bmatrix}$

Attention with position: $q_1 \cdot k_3 = (0.54)(-0.99) + (0.84)(0.14) = -0.41$

Now, relative position difference is 2. What is $\cos 2$? -0.41 !

ie. Attention sees **cos(relative distance)**.

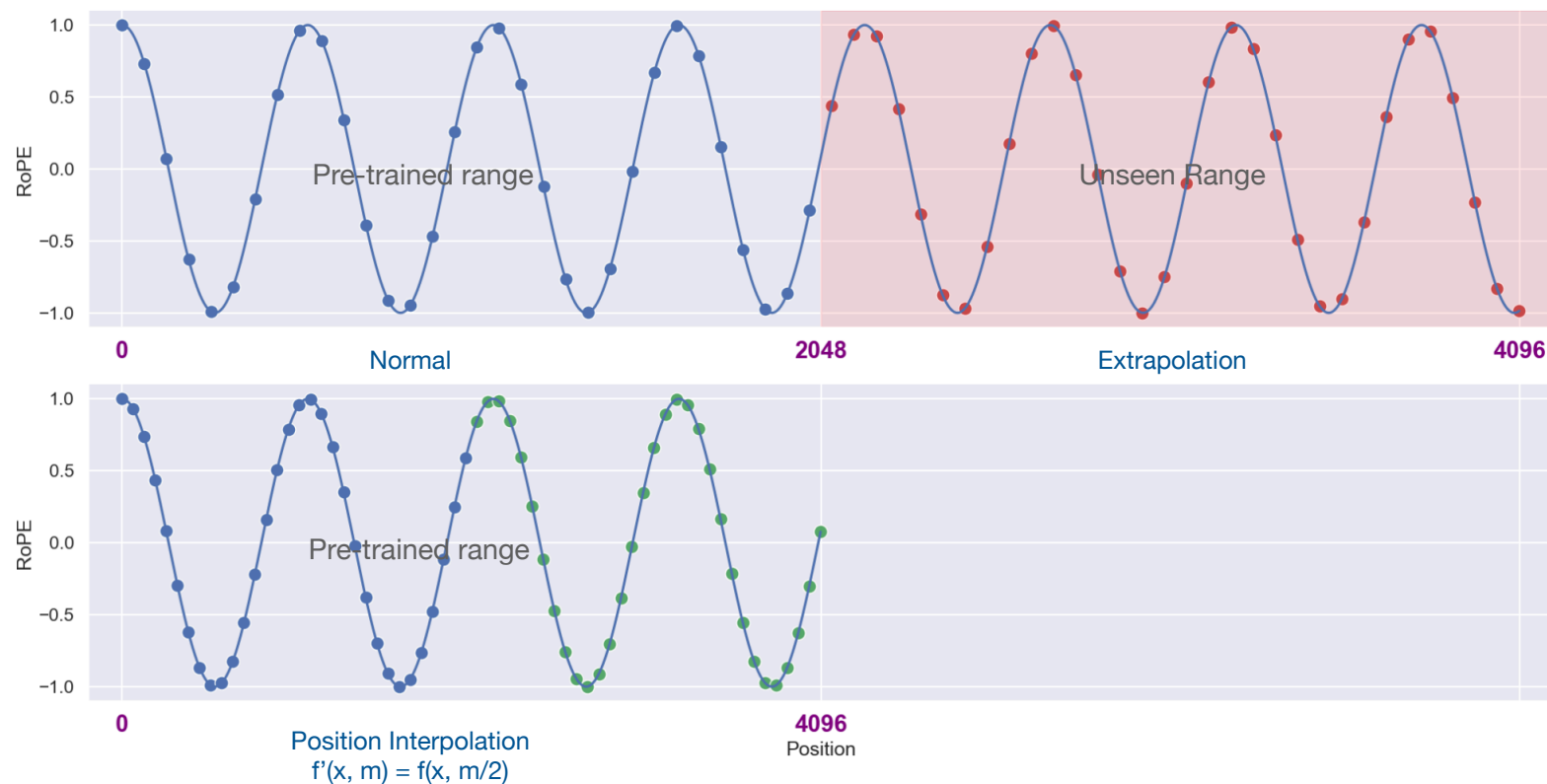
RoPE

- Do not want to transform value V , only attention (thru Q and K)
- Rotation with **sin** and **cos** form an orthogonal transform which is
 - Length-preserving
 - Direction-sensitive
- This gives:
 - Stability
 - Extrapolation
 - No distortion of content magnitude

Model	Validation	Test
BERT-512	64.13%	67.77%
WoBERT-512	64.07%	68.10%
RoFormer-512	64.13%	68.29%
RoFormer-1024	66.07%	69.79%

Extending Context Beyond Training

- Training with long context is expensive!
- And RoPE struggles to generalize past the context window seen during pre-training. Why?
High-frequency rotations wrap around too fast and relative distances become noisy



Position Interpolation:
Extended from 2048 to 32k
Fine tuning on only 1000 steps
Negligible compared to pre-train

Chen et al. (2023) [Extending Context Window of Large Language Models via Positional Interpolation](#)

NTK Scaling

Problem! With Position Interpolation when you compress positions linearly, adjacent tokens become harder to distinguish.

Eg. Original positions: 100, 101, 102, 103

After interpolation 4 x compression: 25.0, 25.25, 25.5, 25.75

NTK (Neural Tangent Kernel – why scale changes learning dynamics – just theoretical connection)
Scaling behaviour of attention – want it to be stable as context length increases

- RoPE rotation angle $\theta = \omega_i \cdot pos$
- RoPE-NTK $\theta = (\alpha \omega_i) \cdot pos$ Where $\alpha < 1$ for longer contexts

Position Interpolation: Compress position indices, keeping frequencies fixed

NTK Scaling: Keep position indices, adjust rotation frequencies

See <https://amaarora.github.io/posts/2025-09-21-rope-context-extension.html>

YaRN

YaRN: Yet another RoPE extension method: PI in parts

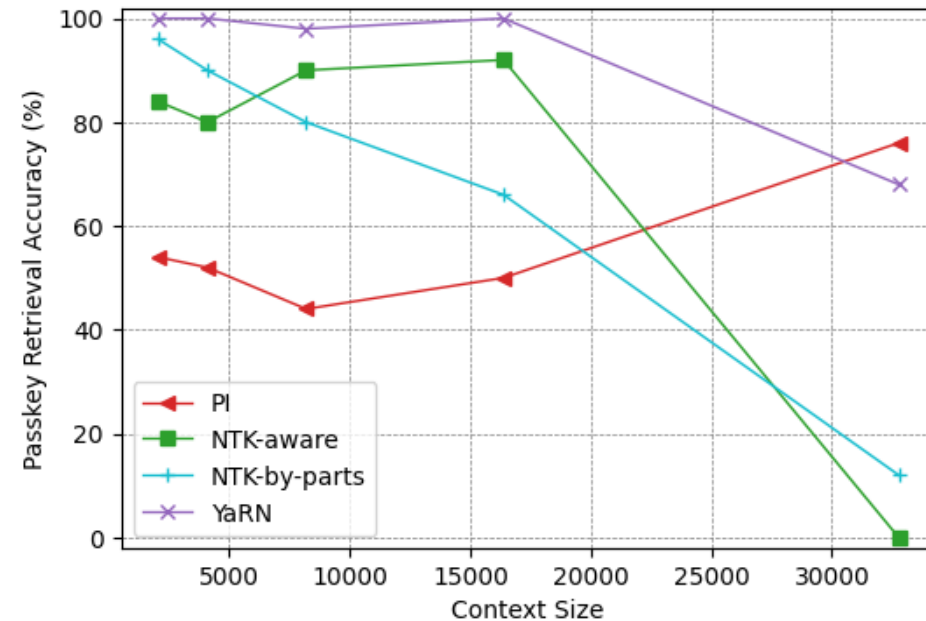
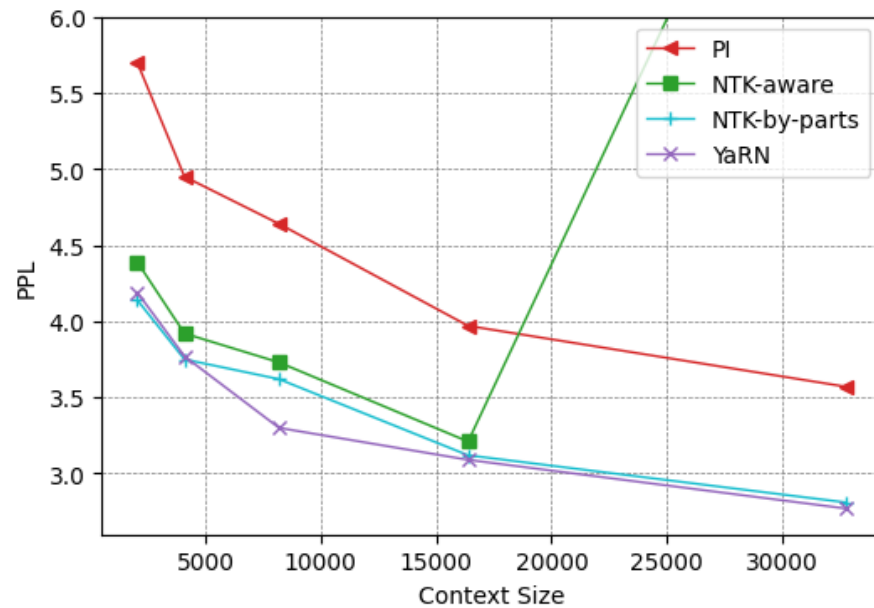
Pretrained Context length = L

Wavelength λ = number of tokens for a RoPE dimension to complete rotation

$\ll L$ no positional interpolation

in between some positional interpolation

$\geq L$ positional interpolation



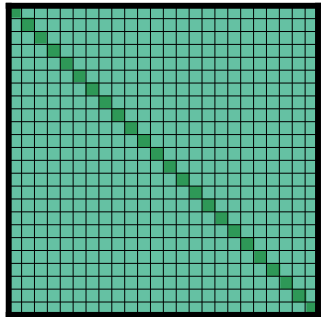
Pushed context > 128k

Peng et al. (2023) [YaRN: Efficient Context Window Extension of Large Language Models](#)

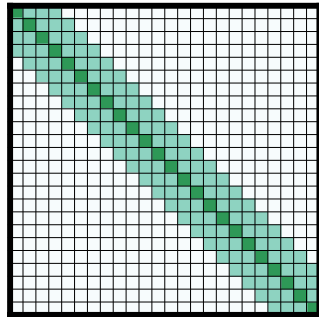
Efficient Attention

Sparse Attention

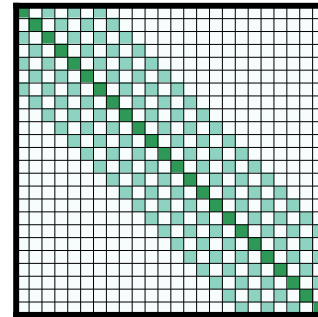
Sparse attention reduces complexity from $O(n^2)$ to $O(n)$



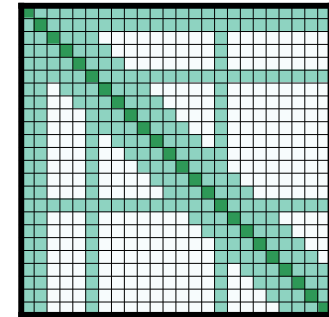
(a) Full n^2 attention



(b) Sliding window attention



(c) Dilated sliding window



(d) Global+sliding window

Showed could process sequences beyond 512 tokens – to 4096

Beltagy et al. (2020) [Longformer: The Long-Document Transformer](#)

Really Long Context

Claude Sonnet 4, Google's Gemini 2.5, GPT-4.1, and Meta's Llama 4 Maverick all offer 1 million token context windows

- Rope scaling methods eg. YaRN
- Efficient attention
- Quantisation
- Importantly requiring minimal fine-tuning – 0.1% of original pretraining data
- However effectively using 100k+ attention remains challenging due to attention degradation over long distances

Evaluation

Needle in the Haystack

Can the model access long context? Simple, controllable sanity check

- The model is given a **very long context** (e.g. 50k–200k tokens).
Most of the text is **irrelevant filler** (the “haystack”): Eg. Wikipedia articles, News text
- Somewhere in the middle (often randomly placed) is a **single critical fact** (the “needle”).
- Ask the model to retrieve this statement
- Iterate over various document depths (where the needle is placed) and context lengths to measure performance

Needle in the Haystack

Input Context

Write a high-quality answer for the given question using only the provided search results (some of which might be irrelevant).

Document [1] (Title: Asian Americans in science and technology) Prize in physics for discovery of the subatomic particle J/ψ . Subrahmanyan Chandrasekhar shared...

Document [2] (Title: List of Nobel laureates in Physics) The first Nobel Prize in Physics was awarded in 1901 to Wilhelm Conrad Röntgen, of Germany, who received...

Document [3] (Title: Scientist) and pursued through a unique method, was essentially in place. Ramón y Cajal won the Nobel Prize in 1906 for his remarkable...

Question: who got the first nobel prize in physics

Answer:

Desired Answer

Wilhelm Conrad Röntgen

Needle in the Haystack

Input Context

Write a high-quality answer for the given question using only the provided search results (some of which might be irrelevant).

Document [1] (Title: List of Nobel laureates in Physics) ...

Document [2] (Title: Asian Americans in science and technology) ...

Document [3] (Title: Scientist) ...

Question: who got the first nobel prize in physics
Answer:

Desired Answer

Wilhelm Conrad Röntgen

Modulating position of needle

Input Context

Write a high-quality answer for the given question using only the provided search results (some of which might be irrelevant).

Document [1] (Title: Asian Americans in science and technology) ...

Document [2] (Title: List of Nobel laureates in Physics) ...

Document [3] (Title: Scientist) ...

Document [4] (Title: Norwegian Americans) ...

Document [5] (Title: Maria Goeppert Mayer) ...

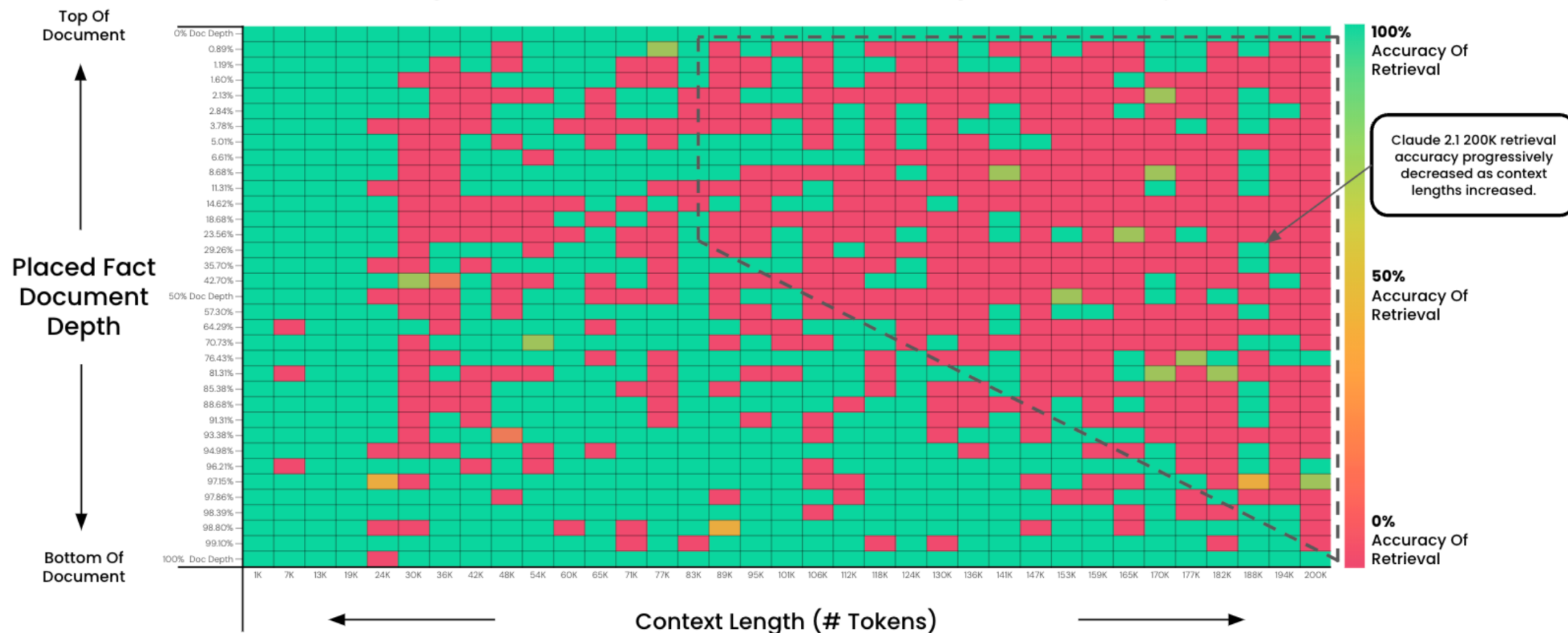
Question: who got the first nobel prize in physics
Answer:

Desired Answer

Wilhelm Conrad Röntgen

Modulating size haystack adding documents

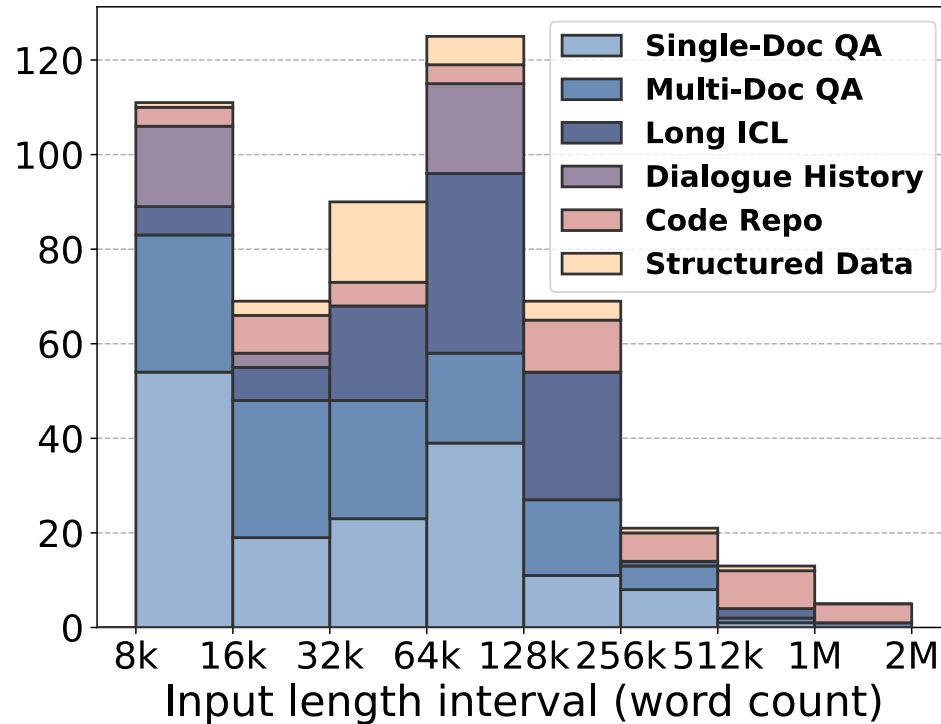
Needle in the Haystack



Claude 2.1: 0-200k token context
35 different document depths top > bottom
35 different context lengths left > right
Evaluated with GPT4 for accuracy

Kamradt et al. (2023) [Needle in a haystack – pressure testing llms](#)

LongBench

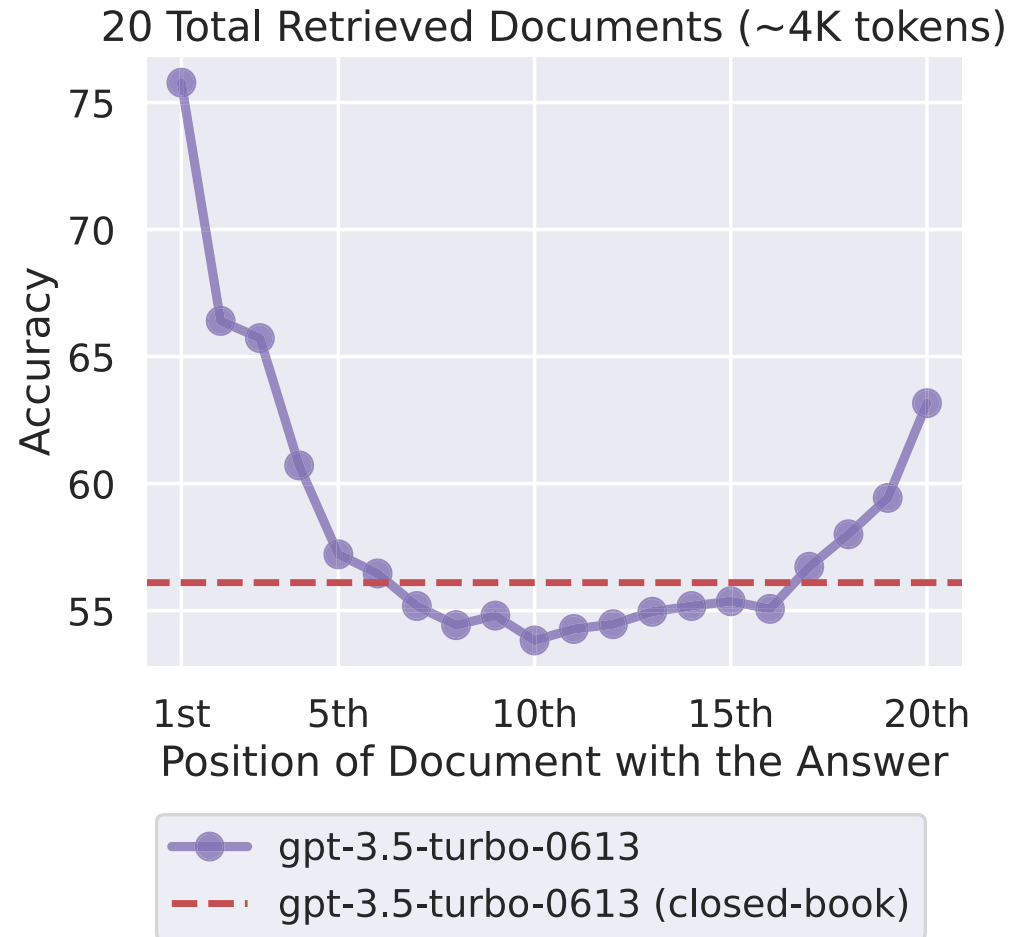


503 challenging MCQ
Context from 8k-2M words
6 major task categories
100 expert annotators
With automated and human review

Bai et al. (2023) [LongBench v2: Towards Deeper Understanding and Reasoning on Realistic Long-context Multitasks](#)

Analysis

Lost in the Middle



Liu et al. (2024) [Lost in the Middle: How Language Models Use Long Contexts](#), ACL

Summary

- Tricky and expensive to get long context tasks to work
- Positional Encodings are key to introducing critical information on sequences
- Huge advances in the last couple of years
- Not clear if using 1M tokens works as well as we think and very expensive
- Still unclear how long context compares to RAG, or context engineering